

Aaron Tran

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Education

University of Waterloo

Waterloo, ON

Bachelor of Computer Science

Sep. 2026 – Apr. 2031

- Relevant Coursework: Designing Functional Programs (CS 135), Algorithm Design & Data Abstraction (CS 136), Algebra (MATH 135), Linear Algebra (MATH 136), Calculus 1 (MATH 137)

Experience

Information Communication Technology Intern

Toronto, ON

Toronto Catholic District School Board

Mar. 2025 – May 2025

- Developed an automated file-renaming program in Python to convert VEX Robotics file formats to STL for 3D modeling and editing workflows, making conversions 15.6x faster than the manual process
- Edited video content and collaborated closely with the ICT department head at St. John Paul II, who led the school's Samsung Solve for Tomorrow initiative, helping the team place in the top 12 regional finalists out of 1,000+ schools nationally in Canada

Projects

Retermina § | *Tauri, Rust, React, TypeScript*

2026 – Present

- Built a desktop terminal workspace on Tauri v2 with a Rust backend driving native PTY shells, running fully local with no cloud dependency or subscription
- Engineered a draggable multi-panel layout with react-grid-layout: split terminals, syntax-highlighted editor, file explorer, live project-wide git diff, localhost tracker, and native preview
- Embedded a Claude Code CLI with per-project token tracking, plus Iris, a context-aware command bar that surfaces git, npm, and shell macros based on live repository state
- Implemented five structural theme engines with portable presets to instantly re-skin the entire application

BAAM § | *Solana, Rust, TypeScript, Swift, MongoDB, Vultr*

JAMHacks, 2026

- Built a social betting platform for close friend groups at JAMHacks, using Solana smart contracts for on-chain wagers, MongoDB Atlas for data, and a Vultr-hosted backend
- Shipped multiple client surfaces: a native iMessage plugin in Swift, a Discord bot, and a centralized web app
- Collaborated with a team to design and deliver the full stack within the hackathon timeframe

FaceRay § | *Python, OpenCV, MediaPipe, Tauri, Rust, TypeScript*

2026

- Built a real-time AI virtual camera that captures a raw webcam feed with OpenCV, extracts a 468-point 3D face mesh via MediaPipe's Face Landmarker, and pipes the processed output into a native OS virtual camera for Discord, Zoom, and Meet
- Implemented eye-contact/gaze correction by anchoring iris landmarks to a configurable attention vector with per-eye temporal smoothing, alongside face-hull-confined filters for skin smoothing, face anonymization, and background blur
- Architected a Tauri 2.0 desktop shell with a strict data/control-plane split: the Python CV core runs as an isolated sidecar so video frames never cross the IPC boundary, while a Rust/TypeScript control panel drives feature toggles over a lightweight stdio JSON protocol
- Streamed processed frames to the live preview over a loopback MJPEG server, keeping heavy frame data off the control-plane IPC

Portfolio § | *Next.js, React, TypeScript*

2026

- Built a personal portfolio site with Next.js, React, and TypeScript using a custom stacked-event architecture for a high-fidelity, animated UI
- Deployed on Vercel for globally distributed performance

Technical Skills

Languages: TypeScript, Python, Rust, Swift, JavaScript, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Tauri, Tailwind CSS, shadcn/ui, OpenCV

Tools & Platforms: Git, MongoDB, Vercel, Vultr, Solana (Anchor)

AI / ML: MediaPipe